Nuclear Engineering and Technology 53 (2021) 1369–1377

Contents lists available at ScienceDirect

Nuclear Engineering and Technology

journal homepage: www.elsevier.com/locate/net

Original Article

Effects of the move towards Gen IV reactors in capacity expansion planning by total generation cost and environmental impact optimization

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ARTICLE INFO

Article history:
Received 11 April 2020
Received in revised form
2 August 2020
Accepted 8 September 2020
Available online 14 September 2020

Keywords:
Expansion planning
Gen IV reactors
WASP

ABSTRACT

Nowadays, it is necessary to accelerate the construction of new power plant in face of rising energy demand in such a way that the electricity will be generated at the lowest cost while reducing emissions caused by that generation. The expansion planning is one of the most important issues in electricity management. Nuclear energy comes forward with the low-carbon technology and increasing competitiveness to expand the share of generated energy by introducing Gen IV reactors. In this paper, the generation expansion planning of these new Gen reactors is investigated using the WASP software. Iran power grid is selected as a case of study. We present a comparison of the twenty-one year perspective on the future with the development of (1) traditional thermal power plants and Gen II reactors, (2) Gen III + reactors with traditional thermal power plants, (3) Gen IV reactors and traditional thermal power plants, (4) Gen III + reactors and the new generation of the thermal power plant, (5) the new generation of thermal power plants and the Gen IV reactors. The results show that the Gen IV reactors have the most developing among other types of power plants leading to reduce the operating costs and emissions. The obtained results show that the use of new Gen of combined cycle power plant and Gen IV reactors make the emissions and cost to be reduced to 16% and 72% of Gen II NPPs and traditional thermal power plants, respectively.

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1. Introduction

Electricity nowadays is being one of the most important energy sectors in any country and plays an important role in the economic decision process with an extreme financial impact [1]. Therefore, the important debate is the issue of generation expansion planning (GEP) to determine which kind and when the generation units should be built over a long-term planning horizon to satisfy the energy demand. Furthermore, it is essential to know the extent of demand for this energy to carefully plan the policy.

Generation Expansion Planning (GEP) is a complex multiyear mixed-integer problem that typically aims at identifying the most adequate schedule for generation investments together with the selection of the locations and technologies to use. Regarding the GEP in particular, the development of electricity markets brought new dimensions to the problem [2]. There are now several generation agents competing to supply the demand. This means that each agent is planning the expansion of own generation assets but it will have to consider the possible behavior of the other competitors and the impact of their expansion plans both in terms of the timing and the technologies. Moreover, the problem is much more contaminated by uncertainties. These uncertainties are related to the long-term evolution of the demand, fuel prices, investment and maintenance prices and also the behavior of the other generation agents, there is some degree of interaction between the decisions of the different generation agents, in the coming years. This ultimately means that rather than optimal expansion long-term plans, generation agents will certainly privilege whenever possible shorter-term robust decisions and strategies. Furthermore, there is a strong dependency on market prices that, at least in the short-term, display a large volatility. These prices are influenced and influence the demand level. These relations show that there is a complex set of dependencies between different parameters and variables suggesting the adoption of a systematic procedure to capture all of them. Finally, there are several objective functions in such problems. One of the mostly

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https://doi.org/10.1016/j.net.2020.09.005
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applied objective function is to diminish emissions and costs.

There are several attempts to find a comprehensive solution to this interesting paradox. The Korean government has established a long-term to supply the demand for electricity by taking in to account the society and economy of the country. They used WASP to develop and evaluate the energy supply and demand plans [3]. The WASP package was originally developed by the Tennessee Valley Authority and the ORNL in the request of IAEA for analyzing the economic competitiveness of nuclear power for supplying forecasting energy demand. The last WASP-IV was released by IAEA for taking into account the fuel availability and environmental constraints. This package determines what kind of power plants should be scheduled to take place to meet demand reliability levels and minimize costs and pollution. Min et al. evaluated the shift of the energy policy toward renewable energy using the optimum generation model to finds the generation mix with the lowest total cost and environmental impact in South Korea. The reliability of the power system was also assessed [4]. The whale optimization algorithm in generation expansion planning was investigated to find a strategy for supplying future energy demand [5]. Several software packages for capacity expansion have become much commercialized and actively implemented in several developing countries. The electricity generation expansion of Tamil Nadu in India was studied by LEAP and Energy PLAN to reduce greenhouse gases emission [6]. To reduce the air pollution in China, the generating units were expanded in multi-regions with interconnecting power transmission lines. The generation and transmission expansions were considered simultaneously to decentralize power generation and use coal power plants in coal-rich regions [7]. The contribution of the electric vehicles and energy storage technologies in the generation mix was considered to provide more flexibility in the power systems of Spain. The optimum generating unit was obtained by the quadric complimentary programming model [8]. In the other study, the authors presented a multistage generation expansion planning model aiming at minimizing both the planning cost and environmental pollution while incorporating large-scale energy storage systems [9]. Das et al. evaluated the ways of expansion with TIMES software for Bangladesh starting from 2015 to 2045. Scenarios consist of different energy carriers such as electricity import, renewable energies, and finally combination renewable and electricity import [10]. And in the other study, the authors tried to minimum the cost of generation expansion planning using the differential algorithm. The first case was the base case scenario, wherein no inclusion of solar plants. The second was the low solar scenario. The third was the high solar scenario. This study evaluates the impacts of solar energy on development [11]. The generation and transmission expansion planning was considered to minimize the investment and operation costs and CO_2 emission with high penetration of renewable energy in Iran power grid [12].

In this paper, we attempt to determine which kind and when the generation units should be built over a long-term planning horizon to satisfy the energy demand by the previous generation of nuclear power plant (NPP), the Gen IV reactor, and the new generation of combined cycle power plants built by General Electric that has high efficiency. The nuclear energy has long provided a large share of electric power. This energy is ideal for dealing with climate change moving closer to a carbon-free future. To continue and expand this energy and have a bright future in a fully competitive business, Gen IV reactors are currently in progress for commercial applications to address unanswered questions about the reliability and safety features, sustainability to make the fuel utilization more efficient, and competitiveness. There are four concepts selected for these reactors including lead fast reactor (LFR), supercritical water-cooled reactor (SCWR), sodium fast reactor (SFR), and gas fast reactor (GFR). In SFRs, sodium is used as a liquid metal coolant for the fast spectrum reactor and the core fuel element is assembled in a hexagonal lattice. The fuel assembly design looks simpler for a reactor core with a fast neutron spectrum. In LFRs, the lead coolant and nitride fuel are used to extend the reactor coolant outlet temperature. The fuel rods are arranged in a bundle to form the fuel assembly that is placed in the reactor core in a hexagonal arrangement. We used the WASP-IV to consider all the parameters of a power plant for minimizing the production costs and to put the emission of pollutants be at the permitted level. The policy to reduce the emission is the use of green and renewable energy sources with penalties for polluting technologies. We present a comparison of the twenty-one year perspective on the future of Iran with development of (1) traditional thermal power plants and Gen II reactors, (2) Gen III + reactors with traditional thermal power plants, (3) Gen IV reactors and traditional thermal power plants, (4) Gen III + reactors and the new generation of the thermal power plant, (5) the new generation of thermal power plants and the Gen IV reactors. The expansion period from the year 2020–2039 is studied in Iran. Iran is decided to construction of 20,000 MW NPP after conducting location and feasibility studies [13].

2. Method description

The WASP program aims to find the best power system development plan for a maximum of 30 years of study. Any possible sequence of production units added to the system is called the development plan or development policy. Development plans are evaluated using either the objective function or the cost function. The cost has been defined as an objective function for the software. The software must provide expansion planning in such a way that the cost is decreased in each scenario, and the constraints such as emissions of carbon and sulfur dioxides, and penalties for loss of load be meet simultaneously. Cost function here is [14]:

$$B_j = \sum_{t=1}^{T} \left[I_{j,t} + F_{j,t} + L_{j,t} + M_{j,t} + Q_{j,t} - S_{j,t} \right] \tag{1}$$

where I is capitalization cost, F is the fuel cost, L denotes the cost of fuel inventory, M is operational and maintenance costs, O is worth of energy haven't sold, S is recycling amounts from initial fuel investment, t is time by year, T is the study period, B_j is the value of the objective function related to the j_mth development plan. The overhead sign indicates the depreciated values over an arbitrary time reference with the depreciation rate i. the optimal development plan is defined as follows:

$$Min(B_j) \quad \forall j \tag{2}$$

In WASP analysis, all the configurations are first determined, and finally, the most optimal one is selected from all available ones. If K_t vector includes all generating units active in year t in a scheme, it should apply to:

$$K_t = K_{t-1} + A_t + U_t - R_t \tag{3}$$

where A_t is the definitive incremental vector of units in year t related to previous development planning, R_t is the unit removal vector in year t is due to the expiry of the unit life, U_t is the vector of production units is the candidate to be added to the system in year t.

The first two vectors are known and the third vector is unknown to be determined. The critical period, p, is defined as the period of the year where the difference between the production capacity and the maximum demand is at the smallest value. Therefore, the

installed capacity, $C(K_{t,p})$, in a critical period of year t should be constrained:

$$Min \leq C(K_{t,p}) \leq Max \tag{4}$$

The reliability of a system in WASP is expressed by the loss of load probability (LOLP). The user must, as a limitation, set a high limit for this LOLP. The LOLP is an index of reliability of power producers showing the amount of load that is not supplied by capacity of power plants. Also, if in a development plan the annual energy demand, E_t, is greater than the annual energy production, G_t, the total cost of the plan will include a penalty of energy not supplied (ENS). This is a function of the energy not supplied, N_t, determined by the user:

$$N_t = E_t - G_t \tag{5}$$

Now we can define the vectors of the objective function as below:

$$I_{j,t} = (1+i)^{-t'} \sum_k [UI_k MW_k] \quad , t' = t + t_0 - 1 \tag{6}$$

$$S_{j,t} = (1+i)^{-T'} \sum_k [\delta_{k,t} UI_k MW_k] \quad , T = T_0 + t_0 \tag{7}$$

where k is the units that is used for the development plan j in year t, UI_k is the investment cost of the kth unit in MW, MW_k is kth unit capacity in MW, $\delta_{k,t}$ is the recovery coefficient related to unit k at the end of the study period, i is the discount rate, t_0 is the number of years between the base year of the depreciation calculation and the first year of the study period. The fuel cost is represented as:

$$F_{j,t} = (1+i)^{-t'-0.5} \sum_{h=1}^{NHYD} \left[\alpha_h \psi_{j,t,h} \right] \tag{8}$$

where α_h is the probability of weather conditions h th and $\psi_{j,t,h}$ is the total fuel cost at h (th) climate and $NHYD$ the total number of weather conditions specified. The fuel inventory cost is obtained as:

$$L_{j,t} = \left[(1+i)^{-t'} - (1+i)^{-T'} \right] \sum_k [UFIC_k MW_k] \tag{9}$$

where $UFIC_k$ is the cost of inventory fuel of the k th unit. The operation and maintenance cost is achieved by:

$$M_{j,t} = (1+i)^{-t'-0.5} \sum_{l=1}^{NUNI} [UFOM_l MW_l + UVOM_l G_{l,t}] \tag{10}$$

where $NUNI$ is the total units in the system in year t, $UFOM_l$ is the fixed maintenance cost of the unit l (\$/MWh), $UVOM_l$ is the variable maintenance cost of the unit l (\$/Mwah), and $G_{l,t}$ is the expected output of unit l in year t which is calculated as the sum of unit energy output in all weather conditions weighted by the probability of those conditions occurring.

$$O_{j,t} = (1+i)^{-t'-0.5} \sum_{h=1}^{NHYD} \left[a + \frac{b}{2} \frac{N_{t,h}}{EA_t} + \frac{c}{3} \left(\frac{N_{t,h}}{EA_t} \right)^2 \right] N_{t,h} \alpha_h \tag{11}$$

where a, b, and c are the constants (\$/KWh) that are given in the input data, $N_{t,h}$ is the amount of energy not provided for weather conditions h in year t, EA_t is energy demand in the system in the year t. The uncertainties in such optimization problem are usually considered by assuming several scenarios in planning.

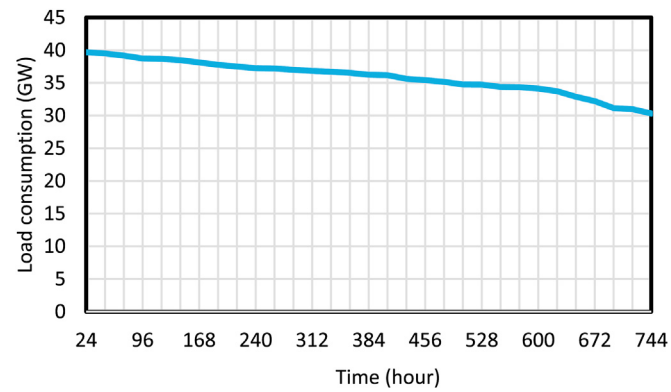

Fig. 1. LDC curve of the first month of the first year of study [16].

3. Electricity energy demand

This section describes the energy demand as input data. The study period from the year 2019 to the year 2039 is analyzed. The load behavior is assumed to have a peak of 57097 MW in the base year [15]. It is also divided into twelve parts each year and the peak load ratio for each month is calculated as a percentage of that year's peak load. The power consumption is assumed to be close to one of the summer months and far from winter months. The daily consumption of each month is also determined by the load duration curve data. The curve is drawn for each day of the month as the vertical axis is the amount of electricity consumed and the horizontal axis is the number of hours associated with that amount of electricity consumed. The graph is plotted point by point for the desired grid. There are thirty days for each month and a certain amount for the consumption of each day. Baseload must be provided throughout the month. The power production should be equal to the base load at least, in all hours. As we move from baseload to peak load, the number of hours becomes less till the day when the peak happens. This is the starting point of the curve. Whatever increases the amount of load, there will be fewer hours to provide the load. For example, Fig. 1 is drawn for the first month of the first year of study.

The peak of load in each year is displayed in Fig. 2. Whatever the assumed Fourier coefficients are chosen more, the accurate software response will be more. The maximum number of these coefficients is 100, which in this study is considered to be 100.

4. Capacity expansion scenarios

The information of the current power plants and candidate power plants will process for development as input. The current plants in the network are divided into five categories: gas, nuclear, gasoline, coal, and mazut burner.

The capacity factor for NPPs is nearly the same as the conventional thermal power plant. Therefore, the reliability of NPPs to deliver power to the loads is approximately the same as conventional power plants. Nevertheless, the risk of NPPs in losing life and polluting environment is much higher than conventional power plant in catastrophic and rare occurring accidents such as that happened in Fukushima. The probability and the consequence of these accident is going to be decreased by use of new generation of NPPs.

Sixty power plants are generally considered to be in the current power system network of Iran, and seventeen parameters for each power plant, including information such as the generation capacity of each power plant, the force outage rate, the contamination rate,

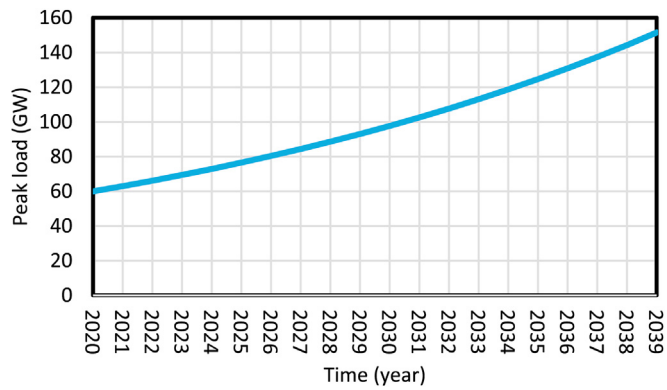

Fig. 2. Peak load information from 2020 to 2039 [17].

the thermal rate, the thermal value, and the fixed and variable operating and maintenance costs and etc. are calculated [18–22] and inputted to the WASP package [22]. Furthermore, all of the thermal power plants are engaged in emission constraints. The pollution gases are assumed to be carbon dioxide and sulfur dioxide. Also, five power plant candidates are considered as follows:

- Traditional thermal power plants and Gen II reactors
- Gen III + reactors with traditional thermal power plants
- Gen IV reactors and traditional thermal power plants
- Gen III + reactors and the new generation of thermal power plant
- The new generation of thermal power plants (ISCC) and the Gen IV reactors.

For the first scenario, gas, steam, combined cycle, coal, and Gen II NPPs are being developed. For the second scenario, gas, steam, combined cycle, coal, and Gen III + NPPs are being developed. The fuel used and the characteristics of the thermal power plants such as gas and steam and combine and coal power plants that running in this scenario, are the same as the fuel and characteristics of the power plants in the first scenario but VVER-1200 and SMART reactors are used as candidate plants for Gen III + NPPs. The VVER-1200 is a large power plant that produce 1200 MWe. The SMART is a modular reactor that produce 100 MWe. We used the large and small power plants to reduce the cost and emissions. The third scenario has the same candidate power plant except using Gen IV NPPs instead of using Gen III + reactors. We applied SFR, LFR, GCR, and SCWR for expansion. In the fourth scenario, the new combined cycle power plant built by General Electric was selected as the new Gen of this power plant. This power plant includes a 330 MW gas turbine and 220 MW steam turbine with a 50 MW solar field capacity with a thermal efficiency of 61%. The third generation of NPPs has been nominated for development in this scenario. In the fifth scenario, the new thermal power plant built by General Electric and fourth-generation NPPs have been nominated for development. In all scenarios, solar photovoltaic (PV), diesel, wind, and hydropower plants are also in development.

All seventeen parameters mentioned, have imported for candidate power plants. And also, to account for operating reserves, between ten and eighty percent of the total production capacity per year is reserved. The details of the candidate development plants are listed in Tables 1–4. Specifications of the new generation of NPPs have been calculated using the G4ECONS and the output of this software has been used as the input of the WASP-IV. G4ECONS was built to calculate the characteristics of new generation plants including the cost of these plants Such as construction costs, operating and maintenance costs, fix and variables costs, fuel costs, etc ….

5. Results and discussion

There are thousands of states for developing in each scenario. All of them must be counted and considered to meet environmental constraints and economic conditions of each candidate power plant like fuel costs, fix and variable costs, and inflation and discount rate. It should be selected to obtain the best expansion plan.

The output of each scenario has been depicted in this section. Some power plants have been involved in all scenarios, like hydropower, PV, and wind power plants and others have only participated in two or more scenarios, like the new generation of NPPs.

Figs. 3 and 4 show the expansion of gas and steam power plants. Figs. 5 and 6 show the expansion of the combined cycle and coal power plants. Figs. 7 and 8 show the expansion of PV and diesel power plants. Figs. 9 and 10 show the expansion of wind and Hydropower power plants. Figs. 11 and 12 show the expansion of old generation nuclear and GCR power plants. Figs. 13 and 14 show the expansion of LFR and SFR power plants. Figs. 15 and 16 show the expansion of SCWR and VVER power plants. Figs. 17 and 18 show the expansion of SMART and ISCC power plants.

The first year of study is 2020 and the last year is 2039. In the first scenario, it is observed that conventional NPP has finally received the most developed among the other plants. There are also PV, diesel, and wind power plants for development in all scenarios. These three distributed generators (DG) compete together for development in all scenarios. The diesel power plant is the last option for development due to the consumption of diesel fuel (gasoline) and higher emissions and is always less developed than other DG(s). The hydropower plant has a capacity of up to 2000 MW depending on the seasons. In the first scenario, the coal power plant is also considered for development. The amount of pollution caused by coal combustion is much higher than that of natural gas combustion. Therefore, the development of these plants has been delayed and the final number is approximately half that of other thermal power plants.

On the other hand, the heat value of coal is lower than the natural gas burned by steam, gas, and combined cycle power plants, which means more fuel is needed and it increases pollution. The combined cycle power plant has priority for development among gas and steam power plants because of its higher efficiency than gas and steam power plants. Of course, the fixed cost of operating a combined cycle power plant is slightly higher than that of gas and steam power plants, but the rate of emission of destructive gases is lower than the gas and steam power plants proportion generation capacity of it.

Traditionally, the wind power grid is cheaper than solar, and from the technical and industrial point of view, the wind power industry can solve many of its challenges, but solar power in the

Table 1
Specifications of hydropower plant [23].

Install (MW)	Storage (MW)	Inflow energy (GWh)	Min. produce (GWh)	Average capacity (MW)	OM cost ($/kW-month)
2000	2400	1100	0	1700	12.3

Table 2
Specifications of fossil and coal-fueled power plants [18–21].

Parameter	Gas	Steam	Combined	Coal	Diesel
Minimum operating level (MWe)	2000	1800	2100	2500	50
Maximum generating capacity (MWe)	2200	2000	2300	2700	55
Heat rate (kcal/kWh)	3300	2900	2048	877	867
Incremental heat rate (kcal/kWh)	2700	2200	1050	700	750
Spinning reserve (%)	10	10	10	10	10
Equivalent forced outage rate (%)	6%	8%	15%	12%	9%
Scheduled maintenance per year (day)	10	10	10	10	0
Maintenance class size (MW)	2200	2000	2300	2700	55
Domestic fuel costs (¢/106 kcal)	3.64	3.78	2.46	3.7	6.6
Foreign fuel costs ($/M kcal)	0	0	0	0	0
Fixed non-fuel OM cost ($/kW-month)	5.7	1.6	2.6	0.05	3.68
Variable non-fuel OM cost ($/MWh)	0.3	0.5	0.3	0.6	0.74
Heat value (kcal/kg)	11000	11000	11100	7950	10986
Emission factor of SO_2 (%)	0.57	5	0.27	1.2	1
Emission factor of CO_2 (%)	0.779	0.63	0.45	1.267	1.044

Table 3
Specifications of power plants [24–28].

Parameter	Gen II NPP	VVER-1200	SMART	PV	Wind
Minimum operating level (MWe)	400	960	20	7	95
Maximum generating capacity (MWe)	967	1200	90.9	10	100
Heat rate (kcal/kWh)	2600	2492.3	2837.7	–	–
Incremental heat rate (kcal/kWh)	2340	2300	2600	–	–
Spinning reserve (%)	10	7	7	5	10
Equivalent forced outage rate (%)	13%	10%	5%	0%	8%
Scheduled maintenance per year (day)	49	20	10	0	5
Maintenance class size (MW)	967	1200	90.9	10	100
Domestic fuel costs (¢/106 kcal)	194	28.61	64.43	0	0
Foreign fuel costs ($/M kcal)	0	0	0	0	0
Fixed non-fuel OM cost ($/kW-month)	8	7.25	6.93	0.04	0.48
Variable non-fuel OM cost ($/MWh)	0.7	0.27	0.258	0	0
Heat value (kcal/kg)	18000	31648	19848.8	–	–
Emission factor of SO_2 (%)	0	0	0	0	0
Emission factor of CO_2 (%)	0	0	0	0	0

Table 4
Specifications of Gen IV NPP [24,29–36].

Parameter	SFR	LFR	GFR	SCWR	ISCC
Minimum operating level (MWe)	800	489.6	864	1281.28	550
Maximum generating capacity (MWe)	1000	599.913	1080	1601.6	550
Heat rate (kcal/kWh)	2262.73	2192.91	1910.76	1919.28	3300
Incremental heat rate (kcal/kWh)	2000	1929.76	1605.29	1842.59	2700
Spinning reserve (%)	7	7	6	7	10
Equivalent forced outage rate (%)	10%	10%	10%	10%	5.40%
Scheduled maintenance per year (day)	30	30	30	30	20
Maintenance class size (MW)	1000	599.91	1080	1601.6	550
Domestic fuel costs (¢/106 kcal)	10.53	51.31	59.07	60.67	3.64
Foreign fuel costs ($/M kcal)	0	0	0	0	0
Fixed non-fuel OM cost ($/kW-month)	7.247	9.469	9.468	9.466	6.917
Variable non-fuel OM cost ($/MWh)	0.27	0.258	0.258	0.258	3
Heat value (kcal/kg)	32138.2	38700	43000	38700	12497.1
Emission factor of SO_2 (%)	0	0	0	0	0
Emission factor of CO_2 (%)	0	0	0	0	3.6073

aspect of the criterion of work, operating worth, energy planning and environmental issues have significant advantages over wind energy. Ultimately, the wind power plant has been more expanded than solar due to its greater capacity than the solar power plant (Figs. 7 and 9). In this scenario, the value of the objective function is 35.49 G$. In other scenarios, it has tried to investigate, the effect of new generation NPPs and the new generation thermal power plants on load supply and the objective function value.

Finally, as seen in Fig. 12 NPP with an approximate capacity of 967 MW is recognized as the first candidate for development due to its zero-emission rate and lower maintenance cost than other plants and has a much higher thermal value than natural gas. The software also allows a maximum of one hydroelectric power plant to be built each year. By enabling the software, it finally identified the optimal construction of the plant in the year 2022 of this Scenario (Fig. 10). Wind and solar power plants are both inexpensive there are no pollutants.

In the second scenario, Gen III + NPPs have the most developing among other power plants due to their heat value and zero rate of emission. VVER heat value is almost three times the other power plants. The G4ECONS was used to obtain the parameters of the new generation of NPPs and the output of this software was used as the

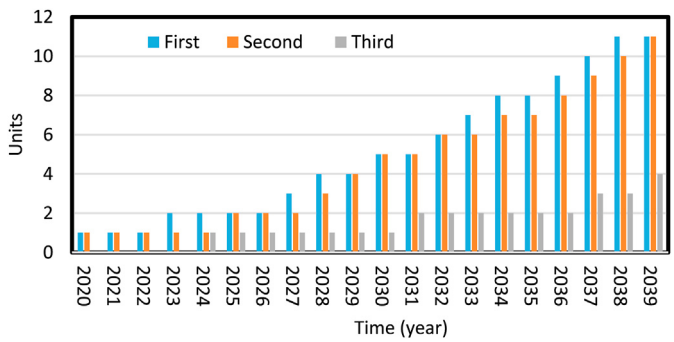

Fig. 3. Development of gas power plant.

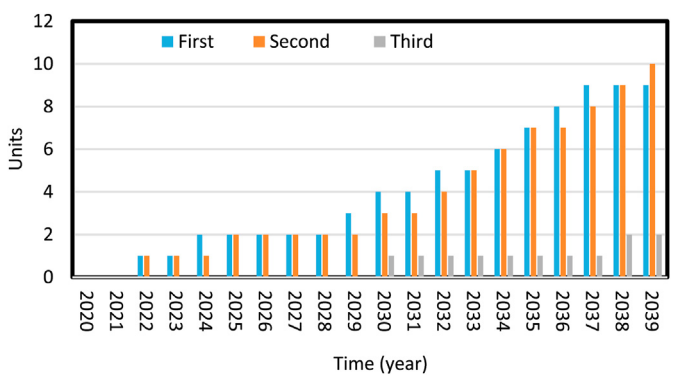

Fig. 4. Development of steam power plant in different scenarios.

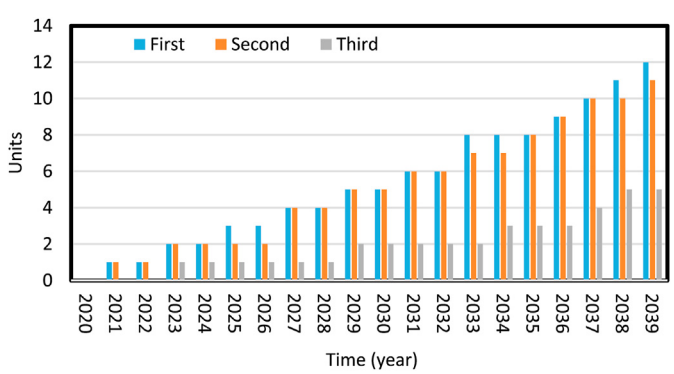

Fig. 5. Expansion of combined cycle power plant in different scenarios.

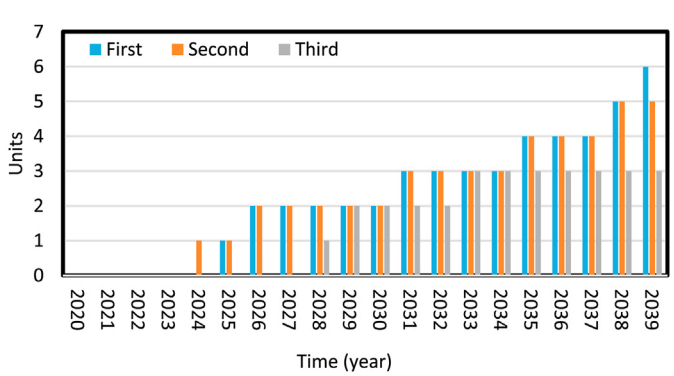

Fig. 6. Expansion of coal power plant in different scenarios.

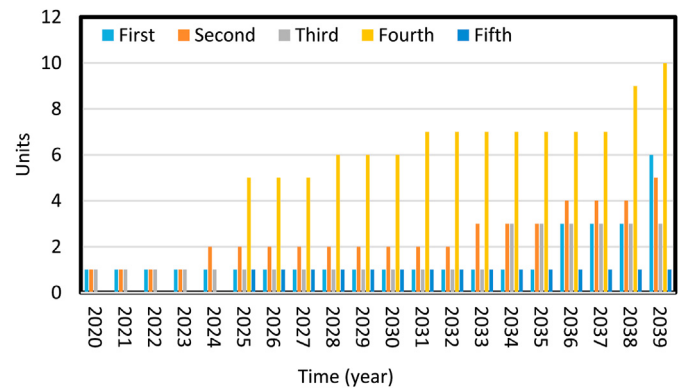

Fig. 7. Expansion of photovoltaic power plant in different scenarios.

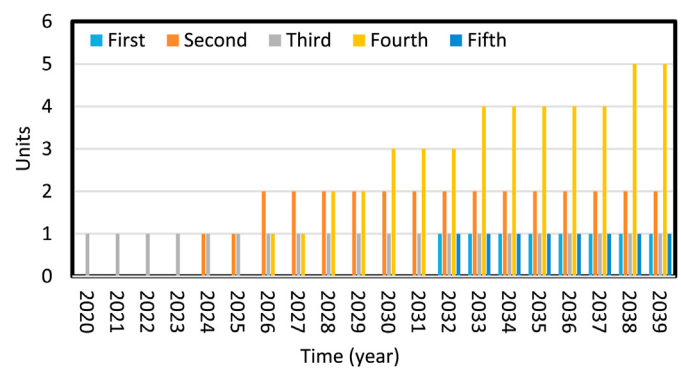

Fig. 8. Expansion of diesel power plant in different scenarios.

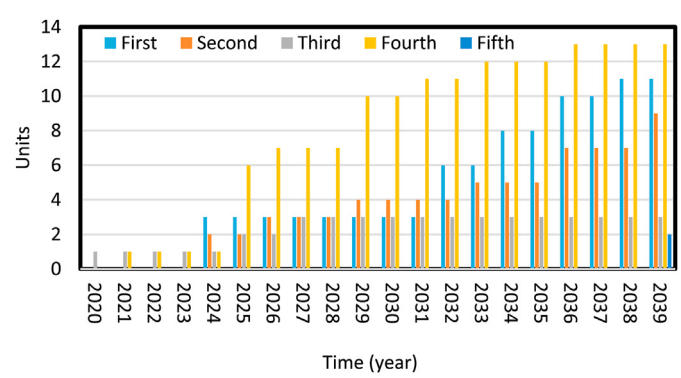

Fig. 9. Expansion of wind power plant in different scenarios.

input of the WASP software. According to the calculations, the variable costs of operating these plants are about the variable costs of thermal power plants, but their fixed costs are about 2.5 times that of the combined cycle power plants. On the other hand, the high thermal value and heat rate of these plants due to reducing costs. Finally, by considering all the positives and negatives of these power plants, the final cost was reduced and the objective function reaches 34.45 G$. It shows using third-generation power plants is economic.

In the third scenario, the fourth generation of NPPs is used. The fixed costs of operating these plants are slightly higher than the third-generation ones. But the thermal value of the power plant is about four times that of the thermal power plant. On the other hand, due to the high technology of these plants, the number of days the plant has to be out of network for repairs is assumed to be

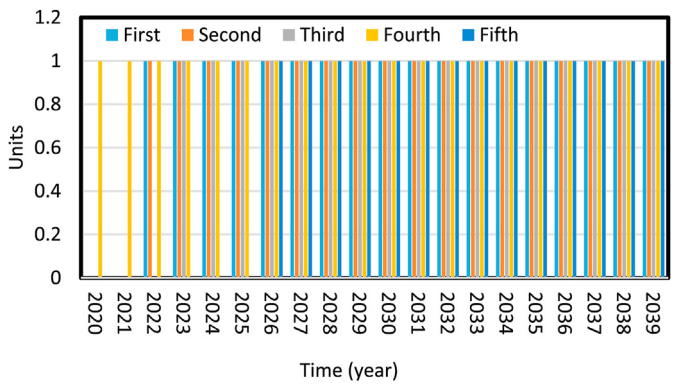

Fig. 10. Expansion of hydropower plant in different scenarios.

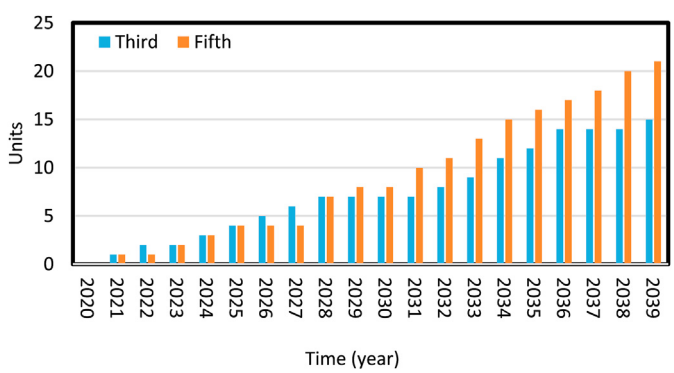

Fig. 11. Expansion of GCR power plant in different scenarios.

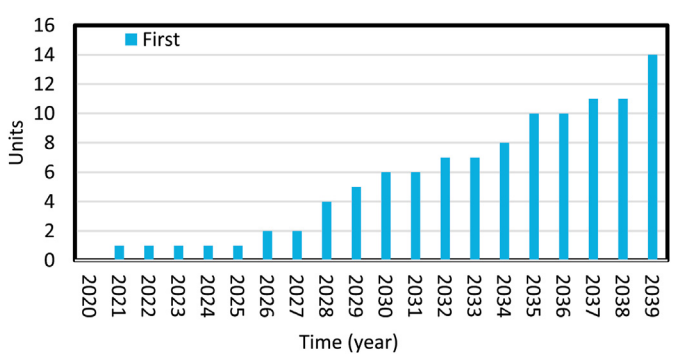

Fig. 12. Expansion of the old generation of the nuclear power plant in different scenarios.

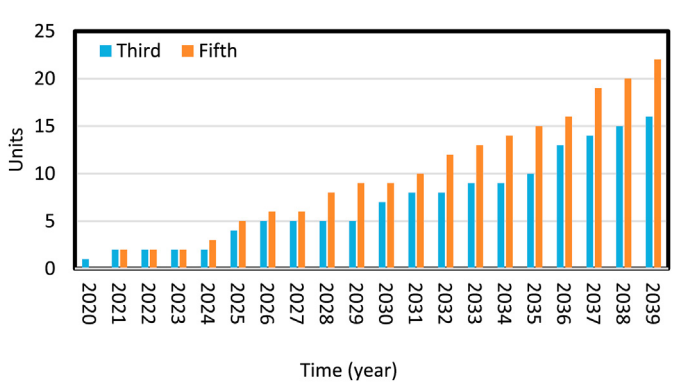

Fig. 13. Expansion of LFR power plant in different scenarios.

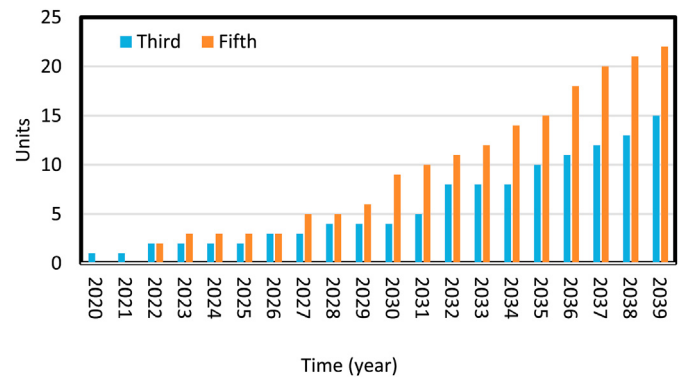

Fig. 14. Expansion of SFR power plant in different scenarios.

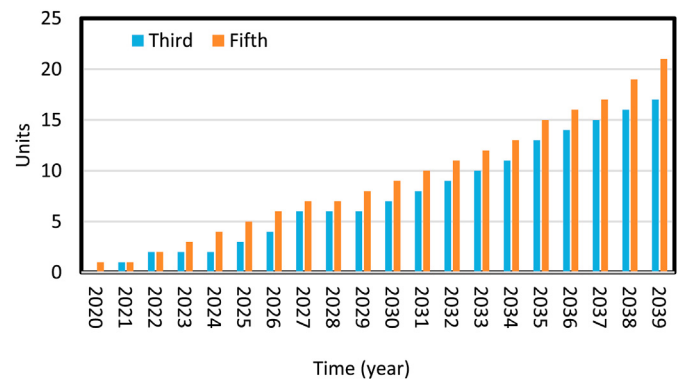

Fig. 15. Expansion of SCWR power plant in different scenarios.

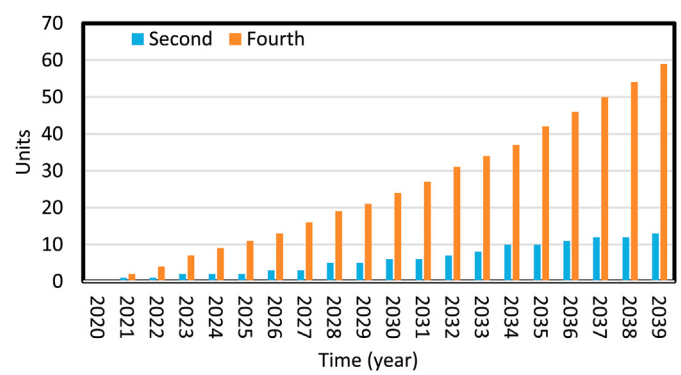

Fig. 16. Expansion of VVER power plant in different scenarios.

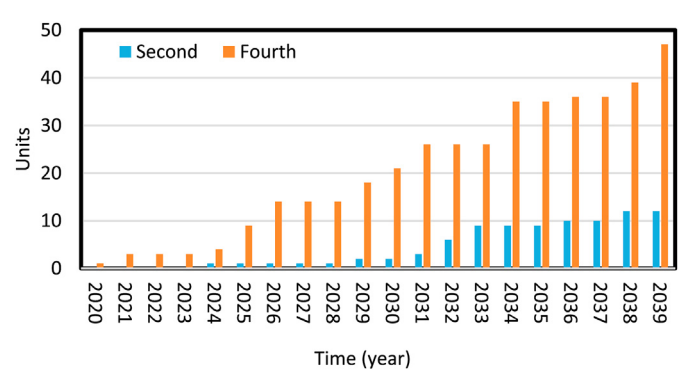

Fig. 17. Expansion of SMART power plant in different scenarios.

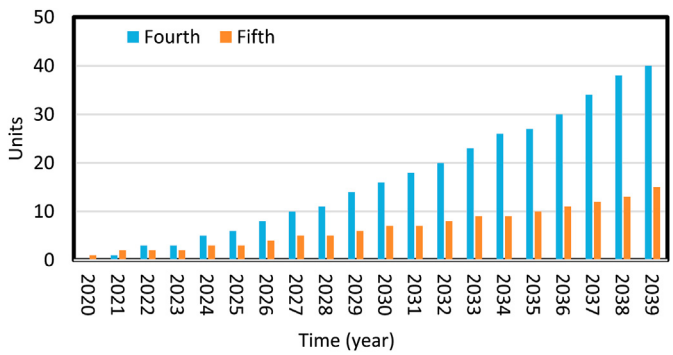

Fig. 18. Expansion of ISCC power plant in different scenarios.

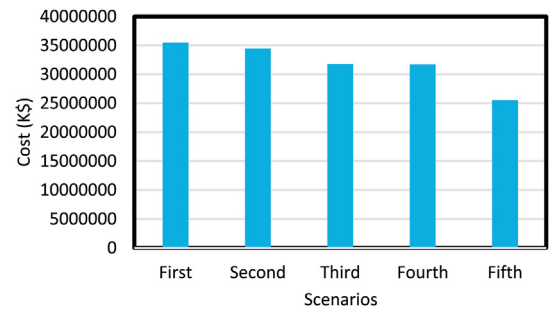

Fig. 19. Cost function during scenarios.

slightly higher than the third-generation plants. The entry of these plants has delayed the development of the hydroelectric power plant and halved the development of the old thermal power plants (Figs. 3–6 and 10). The fact that fourth-generation plants delay the development of hydropower plants is also visible in the fifth scenario. In fact, in the early years of power plant development, fourth-generation plants do not allow any power plants to develop, due to their pollution less and higher efficiency. But ultimately, the high fuel costs of these plants allow the development of other plants. Thus, the construction of this power plant should have greatly reduced the cost of power plant development. The scenario of the development of fourth-generation NPPs has reduced the amount of objective function to 31.75 G$.

In the fourth scenario, the new generation of thermal generators is used instead of all the old thermal power plants. The large number of power plants established in this scenario is that the capacity of the new thermal power plant is assumed to be 600 MW, which can be used in greater numbers and a thermal power plant can consist of several parts. The plant, as mentioned, has high efficiency. The high efficiency of the plant due to the use of the solar power plant during heating states and the high thermal value of the plant compared to a conventional combined cycle power plant will reduce the cost of plant development. On the other hand, in this scenario, third-generation NPPs are used instead of fourth-generation NPPs. As seen in previous scenarios, the use of fourth-generation NPPs instead of third-generation NPPs caused to more reduces development costs. Consider this, in this scenario, the third generation plant is used instead of the fourth generation plant, and the new thermal power plant, is a candidate for development, so the objective function value will remain almost constant. Because the third-generation plants will increase the cost compared to the third scenario (fourth-generation NPPs), but the new thermal power plants will reduce the cost. The objective function in this scenario reduces to 31.69 G$ and almost It has remained constant.

In the fifth scenario, both the fourth generation NPP and the new combined cycle power plant are used. The plan has significantly reduced operating costs. The results also indicate that in competition between the new generation NPPs and the new General Electric thermal power plant, the new generation atomic power plants are superior.in this scenario fourth-generation NPPs have reduced the process of establishing a new generation combined power plant compared to the previous scenario (Fig. 18). The cost of this project is equal to 25.53 G$.

The cost objective function includes capitalization, fuel, operational and maintenance costs and amount of recycled fuel from NPPs and other costs, as discussed in Section 2. Despite having more capital and operational costs for NPPs, the burned fuel can be recycled for reuse and the depreciable capital costs is lower that leads to moderate the plant costs. Furthermore, the penalties and constraints for carbon free electricity generation cannot be applied to NPPs. Thus moving towards the Gen IV NPPs can reduce the costs and emissions. As can be seen, the combination of the new combined cycle power plant with the Gen IV NPPs leads to significant reduction in costs.

For each scenario, the amount of pollutant emissions is calculated. Each scenario, as defined in the power plant specifications, shows the emission of two pollutants. In the first and second scenarios, no significant difference was observed in reducing the amounts of pollutants. The amount of carbon dioxide emissions in both scenarios was 162071 kg at the end of the study periods. Sulfur dioxide emissions were 297 kg in both scenarios at the end of the study periods. But from the third scenario to the fifth scenario is being seen a significant reduction in emission. In third scenario amount of carbon dioxide, reduce to 74934 kg at the end of study periods and for sulfur dioxide, it reached to 98 kg. In fourth scenario amount of carbon dioxide reached to 66685 kg and amount of emission of sulfur dioxide reached to 63 Kg. In the last scenario that is chosen as best plan, the amounts of emissions are least between other scenarios. Carbon dioxide emission reached 26674 kg and sulfur dioxide emission reached 25.3 kg at the end of study periods.

The obtained results show that the use of new Gen of combined cycle power plant and Gen III + reactors make air pollutant emissions is reduced to 41% of Gen II NPPs and traditional thermal power plants. Use of new Gen of combined cycle power plant and Gen IV reactors make the emissions to be reduced to 16% of current Gen II NPPs and traditional thermal power plants. It chooses as best scenario because of impressive reduce in cost and emission.

So all scenarios were implemented and finally, the last scenario is chosen as the best because, by attention to the above discussions and results, it had the lowest cost and lowest emission.

The costs of each scenario are visible in Fig. 19. As it is seen, the objective function is reducing from the first scenario to the fifth scenario.

6. Conclusion

The construction of new power plants should be increased to supply energy demand. To determine which kind and when the generation units should be built over a long-term planning horizon, the generation expansion planning is important. This is a multi-object optimization problem for reducing both the cost and pollution. In this paper, the WASP-IV package to optimize power plant development with five expansion scenarios. The results indicate that the generation cost can be reduced by the use of Gen III + NPPs for 3%; Gen IV NPPs for 11%; Gen III + NPPs with new generation combined cycle power plant for 11%; Gen IV NPPs with new generation combined cycle power plant for 28%. The best plan for power plant development is the use of a fourth-generation NPP and the new General Electric thermal power plant.

Also, it's visible that, in the third scenario, gas power plants establishment become 0.36 time compared with basic scenario, steam power plants also reduce from 9 units to 2 units and have a 78% reduction, combined cycle power plants also have a 41% reduction and reach to 5 units from 12 units, coal power plants also was developed half from 6 units to 3 units. New NPPs even due to reducing the PV and wind power plants. Wind power plants had a 73% reduction.

Using Gen IV NPPs instead of Gen III + NPPs leads to a reduction in planning ISCC power plants from 40 units to 15 units. This scenario is the best scenario for expansion with SFR power plants having 46% growth, LFR power plants having 37.5% growth, GCR power plants having 40% growth and SCWR power plants having 23% growth in comparison to the third scenario. The obtained results has a good consistency with the power plant specifications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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